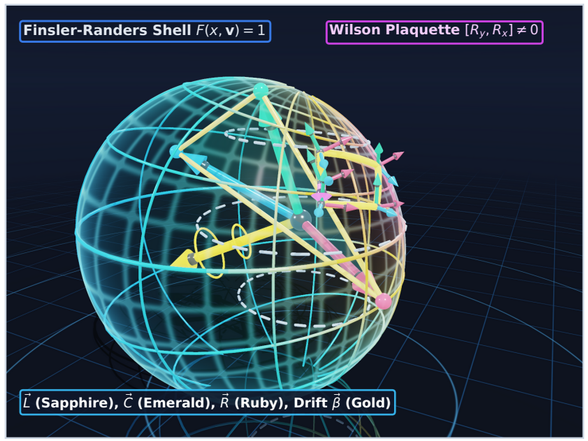
Figure 1 | Non-Abelian 3D Lie Simplex (NA3D), Asymmetric Finsler-Randers Indicatrix & Theorem I Orthogonal Bundle

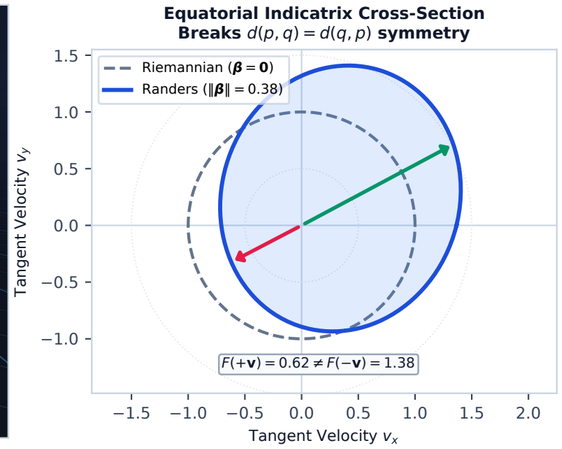
Native Rust GGX 3D ray-traced manifold geometry paired with real intermediate tensors from ARC-AGI-2 Task 1818057f proving zero color-subspace leakage

a 3D NA3D Oriented Simplex & Asymmetric Finsler-Randers Indicatrix Shell

3D Ray-Traced NA3D Simplex & Indicatrix
Asymmetric directional attention modulator

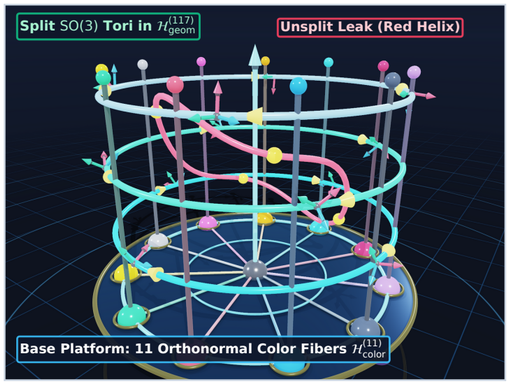


Equatorial Indicatrix Cross-Section
Breaks $d(p,q) = d(q,p)$ symmetry

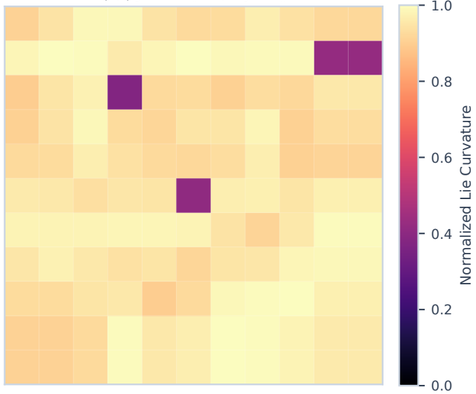


b 3D Orthogonal Fiber Bundle & Plaquette Holonomy Field

3D Ray-Traced Orthogonal Fiber Bundle
Horizontal orbits conserve discrete palette

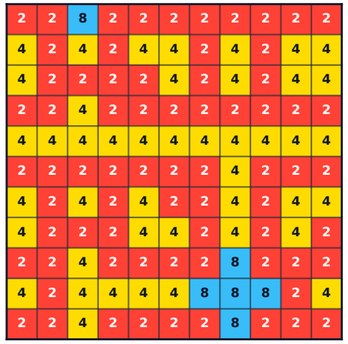


Plaquette Curvature (Task 1818057f)
$\mathcal{K}_{\text{plaq}}(y, x) = \|\nabla_{\text{Lie}}\mathbf{h}(y, x)\|_F$

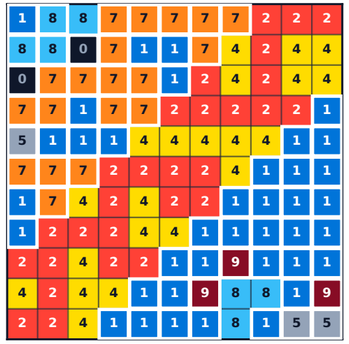


c Visual Proof of Theorem I: Orthogonal Subspace Split Prevents Color Corruption

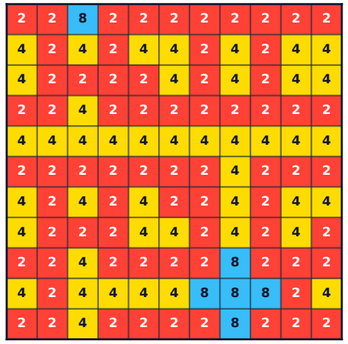
Target Symbolic Patch
Exact palette identities



Unsplit SO(3) Transport
Color-Space Mixing (0.788 Leak)

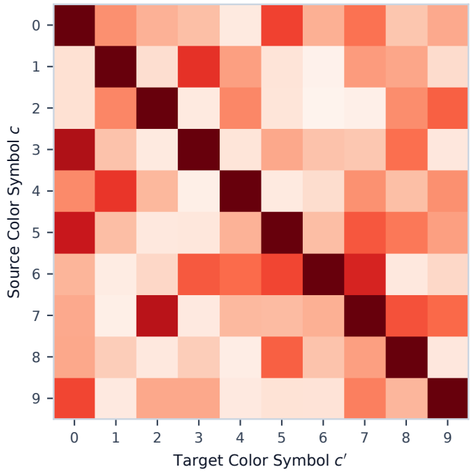


Theorem I Split NA3D
Zero Leak (1.05e-07) | 100% Exact



d Empirical 10×10 Color-Subspace Cross-Leakage Matrix

Unsplit SO(3) Baseline
Frobenius Leakage = 0.788



Orthogonal-Split NA3D (Ours)
Frobenius Leakage = 1.05e-07

